

NICHOLAS LEBEL

+1 (902) 237-0848

lebnico555@gmail.com

[LinkedIn](#)

Education

Dalhousie University

September 2023 – April 2028

Bachelor of Electrical Engineering – GPA: 4.07/4.30

- **Sexton Scholar Recipient** | 2023, 2024 and 2025 | GPA: 4.30, 4.10 and 4.22
Award given to engineering students with a GPA of 3.85 or higher while maintaining a full course load.

Skills

Technical Skills: C, C++, Linux, embedded firmware development, analog/digital circuit design, technical writing.

Frameworks: STM32, AVR, ROS2, Github, MATLAB, KiCad, LTspice, Quartus, Wireshark, Microsoft Suite.

Professional Experience

Lockheed Martin

May 2026 – August 2026

Ship Integration/Test Engineering Co-op

- Translated validation criteria into detailed software test procedures. Executed controlled laboratory testing and reported details to the development team, including initial conditions, edge-case behavior, and anomalous system behavior.
- Conducted shipboard maintenance on the AOPS CCGV, including inspections, repairs and servicing activities related to navigation and combat management systems.
- Designed a Jira-based trial management system for 7 Harbour Acceptance Trials, collaborating with the engineering team to define system requirements. Identified automation opportunities to dynamically track trial progress and update schedules based on metrics such as completion percentage, logged hours, timeline estimates and trial dependencies.

Weir Marine Engineering

May 2025 – September 2025

Application Engineering Intern

- Developed and field-tested a drone control application in C++ which provided functional helm control and waypoint mission planning capabilities to a UAV.
- Conducted test and evaluation trials on RF communications equipment using iPerf to determine network throughput, bandwidth and SNR. The equipment's viability was determined at ranges of up to 1km.
- Participated in field trials involving UAV photogrammetric surveys, UUV sonar mapping and a shock mount trial aboard a Halifax-class frigate.
- Produced technical documentation such as formal reports, SOPs, trial plans/summaries and safety briefs.

Projects

Autonomous Robot

January 2026 – April 2026

Electrical Engineering Design Course

- Designed and implemented an embedded software stack using the AVR ATmega328p, integrating three mission-critical subsystems supporting robot safety and payload retrieval. Includes optical timer-based FSK module, finite state machines and dynamic PWM peripheral control unified into a real-time system deployed during autonomous missions.
- Designed the hardware and software architecture for safety and payload related subsystems. Translated system-level requirements into transistor-based control circuitry for cargo-handling mechanism.
- Integrated the embedded system with a Linux-based companion computer running ROS2. Developed a UART ROS2 topic and ISR to enable exchange of data for payload detection and capture.

Extracurriculars

Dal FSAE

September 2025 – Present

Software Team Lead

- Design and implementation of safety-critical firmware written in C for STM32 devices. Includes CAN network design, IWDG implementation, BMS/HVC and VCU system design, and code reviews to ensure FSAE standards are upheld.
- Responsible for software sprint planning and delegating tasks to members. Involved in leadership and recruitment events.

Canadian Engineering Competition

March 2026

Consulting Engineering

- One of two teams representing Atlantic Canada for Consulting Engineering. Prepared a formal report and presentation in 6 hours proposing a solution to the water management and affordability crisis in Sherbrooke. Placed 8th in Canada.